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EXPERIMENT-1

Aim:- TO PERFORM VERSION CONTROL OPERATIONS USING GITHUB
THIS EXPERIMENT MAPS TO LO2

Git Command	Description
`git config --global user.name "aryarane-459"`	Sets the global Git username used for commits on your system.
`git config --global user.email "aryarane05@gmail.com"`	Sets the global email address associated with your Git commits.
`git add .`	Stages all new, modified, and deleted files in the current directory for the next commit.
`git commit -m "cpp added"`	Creates a commit with the staged changes and the message "cpp added".
`git push origin main`	Pushes the commits from the local main branch to the remote repository (origin).
`git status`	Displays the current state of the working directory and staging area, including modified, staged, and untracked files.
`git branch`	Lists all local branches and highlights the currently active branch.
`Git branch -M main`	Renames the current branch to main.
`git branch feature-branch`	Creates a new branch named feature-branch.
`git switch feature-branch`	Switches from the current branch to feature-branch.
`git checkout -b feature-login`	Creates and switches to a new branch in one command.
`git checkout feature-login`	Switches to the specified branch.
`git push -u origin feature-branch`	Pushes the feature-branch to the remote repository and sets it as the upstream tracking branch.
`git add <file>`	Stages a specific file for the next commit.
`git push`	Pushes the latest commits from the current branch to its configured upstream branch (since the upstream was already set using -u).
`git init`	Initializes a new Git repository in the current directory.

Git Command	Description
`git shortlog`	Displays a summary of commits grouped by author.
`git log`	Shows the complete commit history.
`git log --oneline`	Displays a compact, one-line commit history.
`git diff`	Shows unstaged changes between the working directory and the last commit.
`git fetch origin main`	Downloads updates from the remote repository without merging.
`git branch -a`	Lists both local and remote branches.
`git pull origin main`	Fetches and merges changes from the remote main branch.
`git remote -v`	Displays the remote repository URLs.
`git clone https://github.com/aryarane-459/practicegit.git`	Creates a local copy of an existing remote repository.

Output:

```
@aryarane-459 ➜ /workspaces/practicegit (main) $ git branch
  feature-branch
* main
@aryarane-459 ➜ /workspaces/practicegit (main) $ git switch feature-branch
Switched to branch 'feature-branch'
@aryarane-459 ➜ /workspaces/practicegit (feature-branch) $ git push -u origin feature-branch
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature-branch' on GitHub by visiting:
remote:      https://github.com/aryarane-459/practicegit/pull/new/feature-branch
remote:
To https://github.com/aryarane-459/practicegit
 * [new branch]      feature-branch -> feature-branch
branch 'feature-branch' set up to track 'origin/feature-branch'.
@aryarane-459 ➜ /workspaces/practicegit (feature-branch) $ git status
On branch feature-branch
Your branch is up to date with 'origin/feature-branch'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   quick.cpp

no changes added to commit (use "git add" and/or "git commit -a")
@aryarane-459 ➜ /workspaces/practicegit (feature-branch) $ git add .
@aryarane-459 ➜ /workspaces/practicegit (feature-branch) $ git commit -m "Update quick sort implementation"
[feature-branch 8188737] Update quick sort implementation
 1 file changed, 43 insertions(+), 1 deletion(-)
```

```
@aryarane-459 ➜/workspaces/practicegit (feature-branch) $ git init
Reinitialized existing Git repository in /workspaces/practicegit/.git/
```

```
@aryarane-459 ➜/workspaces/practicegit (main) $ git branch
* main
@aryarane-459 ➜/workspaces/practicegit (main) $ git branch feature-branch
```

```
PS C:\Users\15L\Desktop\githubtrial> git config -- global user.email "aryarane05@gmail.com"
PS C:\Users\15L\Desktop\githubtrial> git config -- global user.name "aryarane-459 "
```

```
@aryarane-459 ➜/workspaces/practicegit (main) $ git add .
@aryarane-459 ➜/workspaces/practicegit (main) $ git commit -m "cpp added"
[main 38a2265] cpp added
 1 file changed, 1 insertion(+)
 create mode 100644 quick.cpp
@aryarane-459 ➜/workspaces/practicegit (main) $ git push origin main
\Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 288 bytes | 288.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/aryarane-459/practicegit
   46e7600..38a2265  main -> main
@aryarane-459 ➜/workspaces/practicegit (main) $ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
@aryarane-459 ➜/workspaces/practicegit (main) $ git branch
* main
@aryarane-459 ➜/workspaces/practicegit (main) $ git branch feature-branch
```

```
@aryarane-459 ➜/workspaces/practicegit (feature-branch) $ git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 671 bytes | 671.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/aryarane-459/practicegit
   38a2265..8188737  feature-branch -> feature-branch
@aryarane-459 ➜/workspaces/practicegit (feature-branch) $ git push
Everything up-to-date
```

```
PS C:\Users\15L\Desktop\githubtrial> git clone https://github.com/aryarane-459/practicegit.git
Cloning into 'practicegit'...
remote: Enumerating objects: 28, done.
remote: Counting objects: 100% (28/28), done.
remote: Compressing objects: 100% (19/19), done.
remote: Total 28 (delta 5), reused 16 (delta 2), pack-reused 0 (from 0)
Receiving objects: 100% (28/28), 5.99 KiB | 2.99 MiB/s, done.
Resolving deltas: 100% (5/5), done.
```

```
@aryarane-459 ➜/workspaces/practicegit (main) $ git remote -v
origin  https://github.com/aryarane-459/practicegit (fetch)
origin  https://github.com/aryarane-459/practicegit (push)
```

```
@aryarane-459 ➜/workspaces/practicegit (main) $ echo "text" > file.py
@aryarane-459 ➜/workspaces/practicegit (main) $ git add experimental.py
fatal: pathspec 'experimental.py' did not match any files
@aryarane-459 ➜/workspaces/practicegit (main) $ git shortlog
aryarane-459 (14):
      Initial commit
      Update README.md
      updated message
      hello
      F1
      Merge pull request #1 from aryarane-459/F1
      F2
      Merge pull request #2 from aryarane-459/F1
      selection sort
      subs
      Merge pull request #3 from aryarane-459/feature
      cpp added
:...skipping...
aryarane-459 (14):
      Initial commit
      Update README.md
      updated message
      hello
      F1
```

```
@aryarane-459 ➜/workspaces/practicegit (main) $ git checkout -b feature-login
Switched to a new branch 'feature-login'
@aryarane-459 ➜/workspaces/practicegit (feature-login) $ git switch main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
```

```
@aryarane-459 ➜/workspaces/practicegit (main) $ git log
Date:   Wed Jul 15 08:26:21 2026 +0000

    Update quick sort implementation
:...skipping...
commit 64c91f387d7910f6fef1ae0362d69babd4482a25 (HEAD -> main, origin/main, origin/HEA
D, feature-login)
Merge: 38a2265 8188737
Author: aryarane-459 <aryarane05@gmail.com>
Date:   Wed Jul 15 13:59:08 2026 +0530

    Merge pull request #4 from aryarane-459/feature-branch

    Update quick sort

commit 8188737ca278da02ec9a4f8895c4c1c953d2e14d (origin/feature-branch)
Author: aryarane-459 <aryarane05@gmail.com>
Date:   Wed Jul 15 08:26:21 2026 +0000
```

```
@aryarane-459 →/workspaces/practicegit (main) $ git log --oneline
64c91f3 (HEAD -> main, origin/main, origin/HEAD, feature-login) Merge pull request #4
from aryarane-459/feature-branch
8188737 (origin/feature-branch) Update quick sort implementation
38a2265 cpp added
46e7600 Merge pull request #3 from aryarane-459/feature
3343fab (origin/feature) subs
73b22ba selection sort
4e0c4a4 Merge pull request #2 from aryarane-459/F1
b28c438 (origin/F1) F2
044b516 Merge pull request #1 from aryarane-459/F1
847a1a3 F1
591a570 hello
6f88f52 updated message
ece2fcb Update README.md
bb5fe21 Initial commit
```

```
@aryarane-459 →/workspaces/practicegit (main) $ git fetch origin main
From https://github.com/aryarane-459/practicegit
 * branch            main       -> FETCH_HEAD
@aryarane-459 →/workspaces/practicegit (main) $ git branch -a
  feature-login
* main
  remotes/origin/F1
  remotes/origin/HEAD -> origin/main
  remotes/origin/feature
  remotes/origin/feature-branch
  remotes/origin/main
@aryarane-459 →/workspaces/practicegit (main) $ git pull origin main
From https://github.com/aryarane-459/practicegit
 * branch            main       -> FETCH_HEAD
Already up to date.
```

```
@aryarane-459 →/workspaces/practicegit (main) $ gh auth status
github.com
  ✓ Logged in to github.com account aryarane-459 (GITHUB_TOKEN)
  - Active account: true
  - Git operations protocol: https
  - Token: ghu_**********************************
@aryarane-459 →/workspaces/practicegit (main) $ gh auth login --scopes "repo"
? Where do you use GitHub? GitHub.com
The value of the GITHUB_TOKEN environment variable is being used for authentication.
To have GitHub CLI store credentials instead, first clear the value from the environme
nt.
```

Conclusion : This experiment provided practical exposure to the core concepts and workflow of Git and GitHub. It involved initializing a Git repository, monitoring repository status, staging and committing changes, creating and switching branches, configuring remote repositories, pushing and pulling updates, reviewing commit history, comparing file changes, restoring files, amending commits, resetting commits, and using the GitHub CLI for authentication. Overall, the experiment strengthened the understanding of version control, efficient branch management, and collaborative software development using Git and GitHub.